MILITARY SPECIFICATION

TEST SET, ELECTRONICS SYSTEM AN/TSM-100

1. SCOPE

1.1 This specification covers one type of Electronic System Test Set, hereinafter referred to as the "system test set" which forms a part of the Vulcan Air Defense System (VADS) Direct Support/General Support System.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-F-13926	Fire Control Material: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Ordnance Material: Packaging of
MIL-T-43435	Tape: Impregnated, Lacing and Tying
MIL-I-45607	Inspection Equipment, Supply and Maintenance of
MIL-C-46357	Computer, Gun Direction; M18 Manufacturing Processes and Inspection of

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MIL-T-60876	Test Set, Servo Amplifier, TS-2658/TSM-100
MIL-T-60877	Test Set, Distribution Box TS-2659/TSM-100
MIL-T-60878	Test Set, Sight Current Generator TS-2660/TSM-100
MIL-T-60879	Test Set, Power Panel TS-2661/ TSM-100
MIL-T-60880	Test Set, Control Panel TS-2657/TSM-100
MIL-T-60885	Test Set, Anti-Aircraft Artillery Gun, AN/MWM-2

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-461	Electromagnetic Interference Characteristics, Requirements for Equipment
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-810	Environmental Test Methods

DRAWINGS

U. S. Army, Frankford Arsenal

F10545638	Test Set, Electronics System AN/TSM-100
F10554035	Schematic Diagram, Electrical, Motor Assembly and Internal Cabling

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F10554070	Drawing Tree for VADS Field Maintenance Test Equipment
C10554215	Power Supply

U. S. Army, Rock Island Arsenal

F8438300	Gun, AA, Towed, 20MM, XM167
F8437450	Gun, AA, Artillery, SP, 20MM, XM163

PACKAGING DATA SHEET

10545638	Packaging of Test Set, Electronics System

(Copies of specifications, standards, drawings and packaging data sheets, required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 Fabrication. The system test set shall be manufactured in accordance with Drawing F10545638 and drawings pertaining thereto.

3.1.1 Details of component parts and subassemblies. Details of design and performance characteristics of all major parts and subassemblies comprising the system test set shall be as shown and as specified on drawings referenced on Drawing Tree F10554070 and specifications listed below. All such parts and subassemblies shall have been inspected and accepted in accordance with their respective drawings and specifications prior to incorporation into the system test set.

Test Set Subassembly	Specification
Servo Amplifier Test Panel	MIL-T-60876
Distribution Box	MIL-T-60877
Sight Current Generator	MIL-T-60878
Power Panel	MIL-T-60879
Control Panel	MIL-T-60880

3.2 General specifications. Where applicable, the system test set shall meet the requirements of MIL-F-13926 and MIL-C-46357 as follows:

3.2.1 MIL-F-13926

a. Order of precedence

b. Dimensions and tolerances

c. Inorganic protective surface finishes

d. Workmanship

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3.2.2 MIL-C-46357

a. Etched circuit assembly. The requirements for etched circuit assemblies of MIL-C-46357 shall apply with modification as follows:

1. Plated etched circuit boards. In addition, assemblies shall not contain copper or nickel visible through hot solder dip, when viewed at 10 times magnification.
2. Location grid. The component mounting holes shall be within 0.007 inch of the true grid intersection, instead of 0.005 inch.
3. Transistor and diode leads. Transistor and metal cased diode leads shall be covered with sleeving in areas where there is a possibility of shorting. No color coding of sleeving shall be required.
4. Drying. The drying process shall not be required.

b. Wiring. The requirements for wiring of MIL-C-46357 shall apply with modification as follows:

1. Spot ties. Tape; Type 1, Finish B per MIL-T-43435 may be utilized without the need for varnish on the knots.
2. Wire termination. Removal of insulation by controlled heat methods shall be acceptable.
3. Chassis cleaning. Compressed air or vacuum may be utilized for cleaning. Vibration shall not be used to clean the chassis.

c. Threaded fasteners. The requirements for threaded fasteners of MIL-C-46357 shall apply with modification as follows:

1. Thread projection. Maximum projection of screws shall be that of the next standard length screw beyond the minimum projection, unless electrical or mechanical interference exists.

d. Marking. The requirements for marking of MIL-C-46357 shall apply with modification as follows:

1. Etching and engraving. Solvo photo etch process is acceptable.

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3.3 Environmental service conditions

3.3.1 Vibration (non-operating). The system test set shall be subjected to a simple harmonic motion having excursions as follows:

Frequency (Hz ± 2 percent)	Excursion (double amplitude) (inches ± 10 percent)
5 to 15	0.06
15 to 25	0.04
25 to 55	0.02

The frequency shall be varied logarithmically between 5 and 55 cycles per second (cps) within 15 minutes and returned to 5 cps in 15 minutes. The motion shall be applied to the system test set for a period of 30 minutes in each direction of the 3 mutually perpendicular axes (total of 90 minutes), relative to its normal operational position. Subsequent to vibration, the system test set shall not be damaged or adversely affected and shall meet the electrical requirements of 3.4.

3.3.2 Temperatures

3.3.2.1 Operating. The system test set shall meet the requirements of 3.4 while exposed to temperatures between +32°F and +131°F.

3.3.2.2 Storage. The system test set shall meet the requirements of 3.4 at temperature of between 60°F and 90°F after having been exposed to a temperature of -80°F and +185°F.

3.3.3 Humidity. The system test set shall function as specified in 3.4 and shall not suffer any damage or deterioration which shall impair satisfactory performance after being subjected to ten consecutive humidity cycles at relative humidities between 85 and 95 percent and temperatures from 82° to 160°F over a period not less than 240 hours.

3.3.4 Shock. The test set shall show no evidence of damage or be adversely affected after being subjected to a total of 18 shocks consisting of three shocks in opposing directions along each of the three mutually perpendicular axes. Each shock shall have an acceleration of 15 gravity units (G's) and a pulse duration of 11±1 millisecond.

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3.3.5 Altitude

3.3.5.1 Operating. The test set shall not be damaged or adversely affected when exposed for a period of not less than four hours to conditions encountered at an altitude of 10,000 feet above sea level. During and subsequent to such exposure the test set shall meet the requirement of 3.4.

3.3.5.2 Non-Operating. The test set shall meet the requirements of 3.4 after exposure to conditions encountered at 50,000 feet above sea level for a period of not less than one hour. The test set shall not be damaged or be adversely affected due to such exposure.

3.3.6 Sand and dust. The equipment shall withstand, in both an operating and non-operating condition, exposure to sand and dust particles as encountered in desert areas.

3.3.7 Salt spray. Equipment shall withstand, in both an operating and non-operating condition, exposure to salt-sea atmosphere.

3.3.8 Fungus-inert materials. The requirements for fungus-inert materials shall be in accordance with MIL-STD-454, Requirement 4.

3.4 Electrical

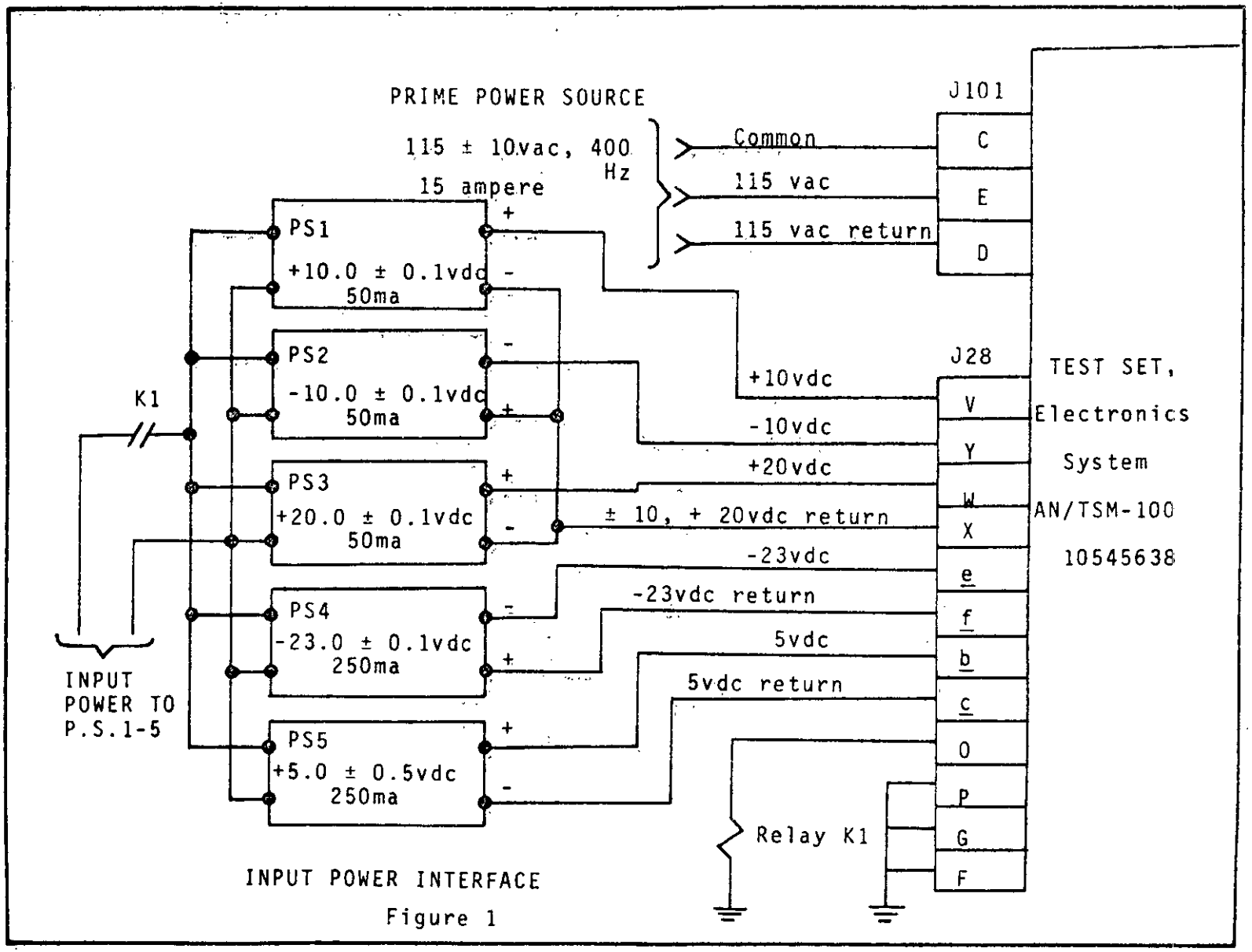
3.4.1 Continuity. The internal cabling and circuitry of the system test set shall be in accordance with electrical schematic Drawing F10554035. The symbols referenced in subsequent paragraphs pertain to this schematic and are used therein.

3.4.2 Conditions of operation. Unless otherwise specified, the requirements of 3.4.3 shall be met under the conditions specified in the following subparagraphs.

3.4.2.1 Input power requirements. The power listed below shall be supplied to the system test set. (See Figure 1)

a. Primary source power
   115 ± 10vac, 400 Hz, 15 ampere
b. Power supplies
   +10.0 ± 0.1vdc, 50 milliamperes
   -10.0 ± 0.1vdc, 50 milliamperes
   +20.0 ± 0.1vdc, 50 milliamperes
   -23.0 ± 0.1vdc, 250 milliamperes
   +5.0 ± 0.5vdc, 250 milliamperes

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INPUT POWER INTERFACE

Figure 1

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3.4.2.2 *Initial switch settings*. Initially, the test set switches and controls shall be placed in the following positions:

a. POWER PANEL

Switch	Position
+24VDC ON-OFF	OFF
115V 400CY 1 PHASE	OFF
DISTRIBUTION +24VDC	OFF
DISTRIBUTION rotary switch	+24 VDC ONLY

b. SIGHT CURRENT GENERATOR

Switch	Position
SYSTEM POWER ON-OFF	OFF
ACTION SWITCH	OFF
FOOT SWITCH	OFF
LOCK ON	OFF

c. DISTRIBUTION

All toggle switches OFF.

d. CONTROL PANEL

+24VDC switch OFF

e. SERVO AMPLIFIER

Switch	Position
+24VDC	OFF
LOAD SELECT	OFF
HAND CONTROL	0 (zero)

f. COUNTER

POWER switch OFF.

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3.4.2.3 Input power interface. The following cable connections shall be made:

(See Figure 1)

CABLE CONNECTOR	CONNECTION
W125P101	To 2J101
W125P27	To POWER SUPPLY, item 2 of 4.4.1.1
W135P1	To J1 of POWER SUPPLY, item 1 of 4.4.1.1
W135P28	To J28 CONNECTOR on test set (2A1)
W136	Between POWER SUPPLIES, items 1 and 2 of 4.4.1.1

3.4.2.4 BATTERY CHARGER initial setup. Prior to performing tests specified herein, the BATTERY CHARGER shall be capable of being set to regulate at a constant rate of +28.5 ± 0.5 vdc when the system test set batteries are being charged at a rate less than 20 amperes.

3.4.3 Functional tests

3.4.3.1 POWER PANEL. With the two DISTRIBUTION, ON-OFF switches on the POWER PANEL in the "ON" position, the +24 VDC indicator lamp and the 115V 400CY 1 PHASE indicator lamp shall light. Also, with the toggle switch on the BATTERY CHARGER in the "ON" position, and the POWER switch on the COUNTER in the "ON (up) position; the AC indicator lamp on the BATTERY CHARGER and the ON indicator lamp on the COUNTER shall LIGHT and 115 ± 10 vac shall be measured between the 115 VAC, + and GRD test points on the POWER PANEL.

3.4.3.2 SIGHT CURRENT GENERATOR. With the DISTRIBUTION switch on the POWER PANEL in the SIGHT CUR GEN position, and with the SYSTEM POWER switch on the SIGHT CURRENT GENERATOR in the "ON" position; the INVERTER indicator lamp and the UNIT NO. 1 indicator lamp on the POWER PANEL shall LIGHT and the SYSTEM POWER indicator lamp on the SIGHT CURRENT GENERATOR shall LIGHT. Also, with the MODE switch of the SIGHT CURRENT GENERATOR in the RADAR READY position; the voltages listed in Table I shall be measured between the test points specified in Table I.

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Table I

Sight Current Generator Test Points Voltages

Test Points		Voltage
+	-	
21	89	-23.0 ± 0.1 vdc
11	88	+20.0 ± 0.1 vdc
12	88	+10.0 ± 0.1 vdc
13	88	-10.0 ± 0.1 vdc
1	2	+18.35± 0.92vdc
6	7	115 v rms ± 10 percent 400 Hz ± 1 percent
3	4	140 v peak to peak ± 16 percent, 400 Hz ± 1 percent quasi-square wave (see Figure 2)
3	5	

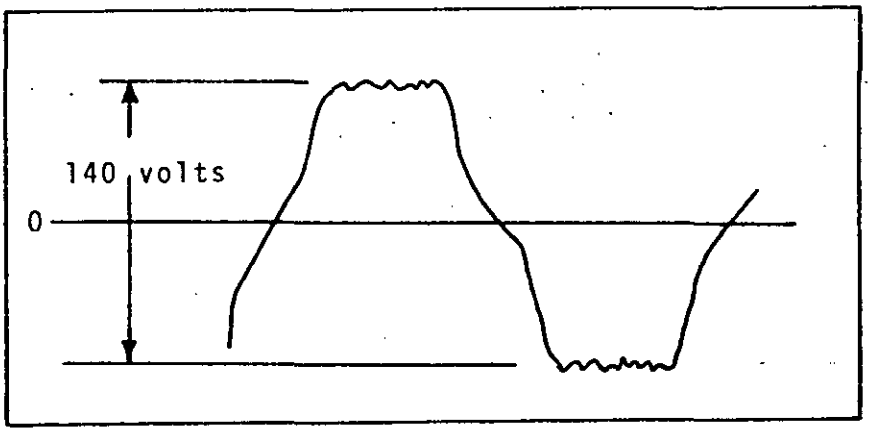


Figure 2

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3.4.3.3 DISTRIBUTION BOX. With the DISTRIBUTION switch on the POWER PANEL in the DB position and with both TEST SYSTEM POWER toggle switches on the DISTRIBUTION BOX in the "ON" position; the UNIT NO. 1 indicator lamp on the POWER PANEL shall LIGHT and the TEST SYSTEM POWER, +5 VDC and +24 VDC indicator lamps in the DISTRIBUTION BOX shall LIGHT.

3.4.3.4 SERVO AMPLIFIER. With the DISTRIBUTION switch on the POWER PANEL in the SVO AMPL position and with the POWER, ON-OFF toggle switch on the SERVO AMPLIFIER in the "ON" position, the POWER, +24 VDC indicator lamp on the SERVO AMPLIFIER shall LIGHT.

3.4.3.5 CONTROL PANEL. With the DISTRIBUTION switch on the POWER PANEL in the CONT PNL position and with the POWER, ON-OFF toggle switch on the CONTROL PANEL in the "ON" position, the POWER, +24 VDC indicator lamp on the CONTROL PANEL shall LIGHT.

3.4.3.6 POWER PANEL, AN/MWM-2. With the DISTRIBUTION switch on the POWER PANEL in the AN/MWM-2 position, the INVERTER indicator lamp on the POWER PANEL shall LIGHT and +15.000 ± 0.075 vdc shall be measured between the 15 VDC, + and GRD test points of the POWER PANEL and -15.000 ± 0.075 vdc shall be measured between the 15VDC, - and GRD test points on the POWER PANEL.

3.4.3.7 POWER PANEL, UNIT NO. 1. With the DISTRIBUTION switch on the POWER PANEL set in the UNIT NO. 1 position; the UNIT NO. 1 indicator lamp on the POWER PANEL shall LIGHT. A potential of +24.0 ± 0.5 vdc shall be applied between Pins J28-C (+) and J28-P (-) of the J28 CONNECTOR located on the side of the test set (2A1) upon which the INVERTER indicator lamp on the POWER PANEL shall LIGHT. And then +24.0 ± 0.5 vdc shall be applied between pins J28-A (+), J28-B (+) and J28-P (-) on the J28 CONNECTOR, the output voltage shall be measured between each of the J28- pins listed in Table II and shall be as specified in Table II.

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Table II

J28 - Output Voltage

J28 Pin		Output Voltage
+	-	
I	H	+15.000 ± 0.075vdc
H	J	-15.000 ± 0.075vdc
L	K	115 v rms ± 10 percent
Z	d	140 v peak-to-peak, quasi-square wave
a	d	140 v peak-to-peak, quasi-square wave

3.4.3.8 POWER PANEL, +24 VDC ONLY. With the DISTRIBUTION switch in the POWER PANEL in any of the +24 VDC ONLY positions, there shall be +25 ± 3vdc measured between pins J27 - Gold (+) and J27 - Silver (-) of the J27 CONNECTOR located on the side of the test set (2A1) and there shall be +25 ± 3 vdc measured between Pins J26 A (+) and J26 B (-) of the J26 CONNECTOR located on the side of the test set (2A1).

3.4.3.9 Motor assembly - Panel 2A2A4

3.4.3.9.1 Counter clockwise output. With the DISTRIBUTION switch on the POWER PANEL in the SVO AMPL position; and with the SERVO AMPLIFIER, DIGITAL VM SELECT switch in the CCW OUPT position; a No. 12 jumper wire connected between Pins J21-M and J21-K; the POWER, ON-OFF switch set in the "ON" position; the SERVO LOAD, LOAD SELECT switch set in the positions specified in Table III; the SERVO LOAD, + and - test points; the TACH, + and - test points and the J25 test point on the SERVO AMPLIFIER shall be monitored and measured voltages shall be as specified in Table III.

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Table III

Servo Amplifier Test Point Voltages

LOAD SELECT switch position	Test Point Voltages		
	SERVO LOAD	TACH*	J25
1/3	11.0 ± 2.0 millivolts	75.0vdc (Min.)	25 ± 2vdc
2/3	25.0 ± 5.0 millivolts	45 ± 5vdc	23 ± 2vdc
FULL	34.0 ± 7.0 millivolts	35 ± 5vdc	22 ± 2vdc
*Note: Tach voltages are negative for CCW rotation and positive for CW rotation.			

3.4.3.9.2 Clockwise output. With the conditions of 3.4.3.9.1 duplicated, the DIGITAL VM SELECT switch shall be set to the CW OUPT position and the No. 12 jumper wire shall be connected between pins J21-L and J21-M. The SERVO LOAD, LOAD SELECT switch shall be set in the Table III positions and the monitored test points voltages shall be as specified in Table III.

3.4.4 Wire (hookup). Stranded conductors of suitable degrees of flexibility shall be used whenever practicable within the requirements of this specification. Hookup wire shall be of such cross section and temper as to provide ample and safe current-carrying capacity and mechanical strength. In general, wire shall not be smaller than AWG No. 22. Wire directly associated with the primary power circuit shall be of a size consistent with the current requirements of the circuit. When it can be demonstrated that the use of smaller hookup wire will result in no loss of performance and at the same time will produce benefits, such as reduction in weight or improved accessibility, smaller wire may be used. This provision is made specifically to permit the use of smaller wire in cables having a large number of wires and having adequate support against adverse effects of vibration. The size of wire leads supplied with parts shall be controlled by the specification for these parts.

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3.4.4.1 Insulation on conductors. Insulation of the sleeve type may be used, provided its length is sufficient to prevent slippage along the conductor to such an extent as to expose a section of the conductor that should be insulated. Insulation shall provide adequate dielectric strength and adequate leakage resistance when the equipment is operated under the designated service conditions. The mechanical and electrical properties of the insulating materials and impregnating compounds shall not be permanently impaired as a result of being exposed to extremes of temperature incurred in the equipment. Rubber without braid may be used only when the insulation is not depended upon to space or support the conductor, or when a number of conductors are assembled in the form of cordage and the assembly provided with a braid covering which has been suitably varnished or lacquered. All conductor insulation shall be of the noncombustible or slow-burning type; that is, with the insulated conductor held in a horizontal position in still air, self-sustained combustion of the insulation or lacquer shall not progress at a rate in excess of 1 inch in 1 minute.

3.4.5 Electromagnetic interference. The test set shall meet the requirements of MIL-STD-461 for type IC equipment; to the extent specified in 6.2 e.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements, as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article (initial production) approval. The requirements for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall be 3 test sets manufactured in the same manner using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing. When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

4.2.2 Contractor testing. When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

4.3 Inspection provisions

4.3.1 Submission of product. Unless otherwise specified herein, or by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" as specified in MIL-STD-105.

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4.3.2 Examinations and tests. Except where otherwise specified herein, components and subassemblies shall be inspected in accordance with the "Inspection provisions for components and subassemblies" and the Sampling Plan Tables contained in MIL-F-13926. Examinations and tests related to Section 3 herein shall be performed in accordance with Tables IV and V and test methods and procedures in 4.5. Examinations and tests for packaging, packing and marking shall be in accordance with MIL-P-14232 and Section 5 herein.

4.3.2.1 Components and subassemblies. Components and subassemblies shall be inspected to the applicable requirements of MIL-F-13926 and MIL-C-46357.

4.3.2.2 Control tests. One system test set shall be selected at random as a control sample from each 100 produced. The control samples shall meet the requirements and tests of Table IV after they have met the requirements and tests of Table V.

Table IV

Test	Requirement	Test Procedures
Fabrication	3.1	Applicable drawings - visual
General specifications	3.2	Visual-tactile
Vibration	3.3.1	4.5.1.1, 4.5.1.2
Temperature	3.3.2	4.5.1.1, 4.5.1.4
Humidity	3.3.3	4.5.1.1, 4.5.1.3
Shock	3.3.4	4.5.1.1, 4.5.1.5
Altitude	3.3.5	4.5.1.1, 4.5.1.6
Sand and dust	3.3.6	4.5.1.1, 4.5.1.7
Salt spray	3.3.7	4.5.1.1, 4.5.1.8
Fungus	3.3.8	4.5.1.1, 4.5.1.9
Electromagnetic interference	3.4.5	4.5.5

4.3.3 Acceptance and rejection. Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and re-submittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

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4.3.4 Functional tests. The requirements and tests in Table V shall be inspected on a 100% basis. Unless otherwise specified, all examinations and tests shall be conducted at the following standard ambient conditions:

a. Temperature $+50^{o}$ to $+90^{o}F$
b. Relative humidity 90% RH max.
c. Pressure. 28 to 32 inches of Hg

Table V

100 Percent Inspection

	Characteristics	Requirements	Procedure
101	Continuity	3.4.1	4.5.2.1
102	Conditions of operation	3.4.2	4.5.2.2
103	Initial switch settings	3.4.2.2	4.5.2.2.1
104	Input power interface	3.4.2.3	4.5.2.2.2
105	BATTERY CHARGER initial setup	3.4.2.4	4.5.2.2.3
	Functional Test		
106	POWER PANEL	3.4.3.1	4.5.3.1
107	SIGHT CURRENT GENERATOR	3.4.3.2	4.5.3.2
108	DISTRIBUTION BOX	3.4.3.3	4.5.3.3
109	SERVO AMPLIFIER	3.4.3.4	4.5.3.4
110	CONTROL PANEL	3.4.3.5	4.5.3.5
111	POWER PANEL, AN/MWM-2	3.4.3.6	4.5.3.6
112	POWER PANEL, UNIT NO. 1	3.4.3.7	4.5.3.7
113	POWER PANEL, +24 VDC ONLY	3.4.3.8	4.5.3.8
114	Motor assembly-Panel 2A2A4	3.4.3.9	4.5.3.9
115	Counterclockwise output	3.4.3.9.1	4.5.3.9.1
116	Clockwise output	3.4.3.9.2	4.5.3.9.2

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4.4 Inspection equipment. Except as otherwise provided for by the contract, the contractor shall supply equipment in accordance with the applicable requirements of MIL-I-45607. The Government reserves the right to use the test equipment for its own independent inspections to the extent that such use will not unduly interfere with the contractor's delivery schedule.

4.4.1 Contractor design. The contractor shall provide inspection equipment compatible with the "Test Methods and Procedures: specified in 4.5 and with the "Component Inspection" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.4.1.1 Test equipment. In conjunction with 4.4.1, the following test equipment shall be utilized in the performance of the applicable tests as specified in 4.5:

Item		Description
1.	POWER SUPPLY	Part Number 10554215, or equivalent
2.	POWER SUPPLY	115 ± 6 vac, 400 Hz, 0-15 amps, single phase, with two output connectors for mating with cable W125P27
3.	DIGITAL VOLT-OHMETER	0.009 to 999 vdc and vac, 0.1% accuracy and resolution. Input impedance shall be greater than 10 megohms on all scales. Resistance accuracy and resolution shall be not less than 0.5% from 1 kilohm to 1,000 kilohms.

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4.	OSCILLOSCOPE	Input impedance greater than 10 megohms, 0 (DC) to 15 megahertz, 5 volts/cm sensitivity; internal sweep, 10 milliseconds/cm and 200 millisecond sweep speeds.
5.	X10 PROBE	Attenuation of 0.1 X input voltage ± 1%. Compatible with item 4.
6.	POWER SUPPLY	24.0 ± 0.5 vdc, 1.5 amperes.
7.	NO. 12 JUMPER WIRE	AWG #12, 10 inches or less.

4.5 Test methods and procedures

4.5.1 Environmental

4.5.1.1 Test conditions. The test set shall be mounted in an upright position in a manner similar to that in which it will be in service and shall be subjected to the environmental tests of the following subparagraphs. The atmospheric pressure shall be room ambient. Except for the high and low temperature tests or the humidity tests, as applicable, the temperature and humidity shall be room ambient.

4.5.1.2 Vibration. The vibration test shall be made on a device with means provided for measuring and controlling the direction of vibration and adjusting the amplitudes and frequencies to keep them within the specified limits of 3.3.1. The measuring instruments of the testing device shall be capable of measuring the vibration amplitude within plus or minus 5 percent, and capable of measuring the frequency within plus or minus 2 percent. The test shall be performed in accordance with the requirements specified in 3.3.1. Subsequent to the vibration test, the test set shall show no damage or defect resulting from the test when examined by visual and tactile means. The test set shall then be inspected to determine compliance with requirements specified under 3.4.

4.5.1.3 Humidity. The humidity test shall be performed in accordance with Procedure I of Method 507 in MIL-STD-810.

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4.5.1.4 Temperature. Test in accordance with Procedure I of Method 501 in MIL-STD-810 for high temperature operating and non-operating under conditions specified in 3.3.2. Test in accordance with Procedure I of Method 502 for low temperature operating and non-operating under conditions specified in 3.3.2.

4.5.1.5 Shock. The shock test shall be performed in accordance with MIL-T-5422.

4.5.1.6 Altitude. The altitude test shall be performed in accordance with the following steps.

Step 1. Place the test set in the test chamber in a manner that will simulate service usage, making connections and attaching instrumentation as necessary. Plugs, covers, and inspection plates not used in operation, but used in servicing shall remain in place. When mechanical or electrical connections are not used, the connections normally protected in service shall be adequately covered. For tests where temperature values are controlled, the test chamber shall be at standard ambient conditions when the test set is installed. The test set shall then be operated to determine that no malfunction or damage was caused due to faulty installation or handling. The requirement for operation following installation of the test set in the test facility is applicable only when operation is required during or immediately following exposure to the specified test. Standard ambient temperature shall be maintained during the entire test.
Step 2. Reduce the chamber internal pressure to 87.5 mm of Hg (3.44 in. of Hg or 50,000 ft. above sea level). The rate of pressure change may be the maximum attainable by the chamber. Maintain the chamber pressure for a period of not less than one hour.
Step 3. Increase the chamber pressure to 523 mm of Hg (20.6 in. of Hg or 10,000 ft. above sea level) and then operate the test set and compare the results with the data obtained in accordance with pretest performance record made of all data necessary to determine compliance with required performance.

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Step 4. With the test set not operating, return the chamber to standard ambient conditions.
Step 5. Operate the test set and compare the data obtained with the data obtained in accordance with pretest performance record.
Step 6. The test set shall be visually inspected and a record made of any damage or deterioration resulting from the test. If a test chamber is used for the test, perform a visual inspection of the test within the chamber at test conditions, when possible. Upon completion of the test, visually inspect the test set again after the test set has been returned to standard ambient conditions. Deterioration, corrosion, or change in tolerance limits of any internal or external parts which in any manner prevent the test set from meeting operational service or maintenance requirements shall provide reason to consider the test set as having failed to withstand the conditions of the test.

4.5.1.7 Sand and dust. The sand and dust test shall be performed in accordance with MIL-T-5422.

4.5.1.8 Salt spray. The salt spray test shall be performed in accordance with MIL-T-5422.

4.5.1.9 Fungus. The fungus resistance test shall be performed in accordance with MIL-T-5422.

4.5.2 Electrical tests

4.5.2.1 Continuity test. A continuity test will be performed to verify conformance with 3.4.1.

4.5.2.2 Conditions of operation. Unless otherwise specified herein, tests of subsequent paragraphs will be performed with the conditions of operation specified in 3.4.2.

4.5.2.2.1 Initial switch settings. Verify that the system test set initial switch settings conform to the requirements specified in 3.4.2.2.

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4.5.2.2.2 Input power interface. With the requirements of 3.4.2.2 met, perform the following interconnections to provide the power as specified in 3.4.2.1. Connect cable connector W125P101 to 2J101. Connect W125P27 to the POWER SUPPLY, item 2 of 4.4.1.1. Connect cable connector W135P1 to J1 on POWER SUPPLY, item 1 of 4.4.1.1. Connect cable connector W135P28 to J28 on the test set. Connect the POWER SUPPLY, item 1, to POWER SUPPLY, item 2, using cable W136 of item 1.

4.5.2.2.3 BATTERY CHARGER initial setup. Verify the requirements of 3.4.2.4 as follows:

a. Remove the test set two front panels, 2A1A5 and 2A2A5. Disconnect P113 and P118 from the two batteries.
b. Rotate the BATTERY CHARGER, CURRENT MODE control to the fully clockwise position. Rotate the VOLTAGE MODE control to the fully counter clockwise position. Connect the DVM, item 3 of 4.4.1.1, to the +24 VDC test points on the POWER PANEL.
c. Set the +24 VDC and 115V 400CY 1 PHASE, ON-OFF switches on the POWER PANEL and the BATTERY CHARGER, ON-OFF switch in the "ON" position. Rotate the BATTERY CHARGER, VOLTAGE MODE control until the DVM indicates +28.5 ± 0.5 vdc. Turn off the BATTERY CHARGER power.
d. Reconnect P113 and P118 to their respective battery. Set the BATTERY CHARGER power on-off switch in the "ON" position. Do not readjust the VOLTAGE MODE control even though the indicated voltage is less than 28.5 VDC. The test set batteries will be charged until the BATTERY CHARGER voltmeter indicates 28.5 ± 0.5 VDC and the VOLTAGE MODE indicator lamp LIGHTS.

4.5.3 Functional test

4.5.3.1 POWER PANEL. Place the two DISTRIBUTION, ON-OFF switches in the POWER PANEL in the "ON" position. Place the BATTERY CHARGER toggle switch in the "ON" position. Place the POWER switch on the COUNTER in the "ON" (up) position. Connect the DVM, item 3 of 4.4.1.1 between the 115 VAC, + and GRD test points on the POWER PANEL. Verify requirements of 3.4.3.1.

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4.5.3.2 SIGHT CURRENT GENERATOR. Place the DISTRIBUTION switch on the POWER PANEL in the SIGHT CUR GEN position. Place the SYSTEM POWER switch on the SIGHT CURRENT GENERATOR in the "ON" position. Place the SIGHT CURRENT GENERATOR, MODE switch in the RADAR READY position. Connect the DVM, item 3 of 4.4.1.1, and the OSCILLOSCOPE with attached X10 PROBE, items 4 and 5 of 4.4.1.1, in turn between each of the TEST POINTS specified in Table I. Verify requirements of 3.4.3.2.

4.5.3.3 DISTRIBUTION BOX. Place the DISTRIBUTION switch on the POWER PANEL in the DB position. Place both the TEST SYSTEM POWER toggle switches in the "ON" positions. Verify requirements of 3.4.3.3.

4.5.3.4 SERVO AMPLIFIER. Place the DISTRIBUTION switch on the POWER PANEL in the CONT PNL position. Place the POWER, ON-OFF toggle switch on the CONTROL PANEL in the "ON" position. Verify requirements of 3.4.3.4.

4.5.3.5 CONTROL PANEL. Place the DISTRIBUTION switch on the POWER PANEL in the CONT PNL position. Place the POWER, ON-OFF toggle switch on the CONTROL PANEL in the "ON" position. Verify requirements of 3.4.3.5.

4.5.3.6 POWER PANEL, AN/MWM-2. Place the DISTRIBUTION switch on the POWER PANEL in the AN/MWM-2 position. Connect the DVM, item 3 of 4.4.1.1, between the 15 VDC, + and GRD test points on the POWER PANEL and measure voltage. Remove the positive connector of the DVM from the 15 VDC + test point and reconnect to the 15 VDC - test point. Measure voltage. Verify requirements of 3.4.3.6.

4.5.3.7 POWER PANEL UNIT NO. 1. Place the DISTRIBUTION switch on the POWER PANEL in the UNIT NO. 1 position. Verify UNIT NO. 1 indicator lamp on the POWER PANEL is LIT. Connect POWER SUPPLY, item 6 of 4.4.1.1, between PINS J28-C (+) and J28-P (-) of the J28 CONNECTOR and apply +24.0 ± 0.5 vdc. Verify the INVERTER indicator lamp on the CONTROL PANEL is LIT. In addition, connection shall be made from the positive terminals of the POWER SUPPLY to PINS J28-A (+) and J28-B (+) on the J28 CONNECTOR and +24 ± 0.5 vdc shall be applied. Utilizing the DVM, item 3 of 4.4.1.1, measure the voltages between each of the J28 Pins listed in Table II and verify the Table II output voltages of 3.4.3.7.

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4.5.3.8 POWER PANEL, +24 VDC ONLY. Place the DISTRIBUTION switch on the POWER PANEL in the +24 VDC ONLY position. Connect the DVM, item 3 of 4.4.1.1, between Pins J27 - Gold (+) and J27 - Silver (-) of the J27 CONNECTOR. Connect the DVM, item 3 of 4.4.1.1, between Pins J26A (+) and J26B (-). Verify voltages measured conform with requirements of 3.4.3.8.

4.5.3.9 MOTOR ASSEMBLY

4.5.3.9.1 Counterclockwise output. Connect CABLE CONNECTOR W101P21 to CONNECTOR J21 on the SERVO AMPLIFIER. CONNECT the No. 12 jumper wire between Pin M and Pin K of the W101P1 CABLE CONNECTOR. Place the DISTRIBUTION switch on the POWER PANEL in the SVO AMPL position. Place the DIGITAL VM SELECT switch on the SERVO AMPLIFIER in the CCW OUPT position. Place the POWER, ON-OFF switch on the SERVO AMPLIFIER in the "ON" position. Connect the DVM, item 3 of 4.4.1.1 between: the SERVO LOAD, + and - test points; the TACH, + and - test points; and the J25 test point on the SERVO AMPLIFIER. Place the LOAD SELECT switch in each of the positions specified in Table III in turn. Measure voltages and verify conformance with Table III of 3.4.3.9.1.

4.5.3.9.2 Clockwise output. Clockwise output testing shall be as specified in 4.5.2.3.10.1 except: the DIGITAL VM SELECT switch shall be placed in the CW OUPT position; and the No. 12 jumper wire shall be connected between Pin M and Pin L of the W101P1 CABLE CONNECTOR. Measured voltages shall conform to the requirements of 3.4.3.9.2.

4.5.4 Wire insulation. Wire insulation requirements per 3.4.4 shall be verified by certification.

4.5.5 Electromagnetic interference. Verification of EMI requirements referenced in 3.4.5 and as specified in the contract or purchase order (6.2 e), shall be in accordance with the requirements of MIL-STD-462.

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5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking shall be in accordance with Packaging Data Sheet 10545638.

6. NOTES

6.1 Intended use. The system test set is a bulkhead-mounted-electrical-electronic console that consists of subassemblies (drawers and panels) referenced in 3.1.1 and called out on Drawing F10545638. The system test set mounts in a mobile van or shop facility by securing to the bulkhead, and is used to troubleshoot and verify performance of four components on VADS equipments. The four components are the Sight Current Generator, Distribution Box, Control Assembly, and Servo Amplifier. The VADS equipments are the XM163 20 MM SP Artillery AA Gun, Drawing F8437450, and XM167 20 MM Towed Gun, Drawing F8438300. The test set also supplies the required electrical power for checkout and troubleshooting the AN/MWM-2 Organization Test Set MIL-T-60885 and the External Range Control Assembly.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Applicable stock number.
c. A block of serial numbers to identify each test set.
d. Instructions for first article inspection.
e. Detailed electromagnetic interference requirements (see 3.4.5).

Custodian:

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Preparing Activity:

Army - MU

Project No. 4931-A008

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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